NILPOTENT ELEMENTS BEHAVE AS EXPECTED UNDER HENSELIZATION AND STRICT HENSELIZATION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The extension of the nilradical along a ring map consists of nilpotent elements. Algebra, Lemma \ref{algebra-lemma-locally-nilpotent}, cited at proof lines 11096-11097. Shows one inclusion.
- The henselization of a quotient is the quotient of the henselization by the extended ideal. Algebra, Lemma \ref{algebra-lemma-quotient-henselization} and Algebra, Lemma \ref{algebra-lemma-quotient-strict-henselization}, cited at proof lines 11098-11099 and 11106. Identifies the quotient of the henselization by the extended nilradical.
- The henselization of a reduced ring is reduced. Lemma \ref{lemma-henselization-reduced}, cited at proof lines 11100-11101. Makes that quotient reduced, giving the reverse inclusion.

Source reference. Original lemma 0ASE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13577–13577. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0544: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6627–6627. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05WQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7594–7594. Navigation pointer to a statement the

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Source reference. Original lemma 06DH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8181–8181. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05WS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7596–7596. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a local ring. Let $nil(R)$ denote the ideal of nilpotent elements of R. Then $nil(R)R^h = nil(R^h)$ and $nil(R)R^{sh} = nil(R^{sh})$.*

Proof. Note that $nil(R)$ is the biggest ideal consisting of nilpotent elements such that the quotient $R/nil(R)$ is reduced. Note that $nil(R)R^h$ consists of nilpotent elements by Algebra, Lemma 0544. Also, note that $R^h/nil(R)R^h$ is the henselization of $R/nil(R)$ by Algebra, Lemma 05WQ. Hence $R^h/nil(R)R^h$ is reduced by Lemma 06DH. We conclude that $nil(R)R^h = nil(R^h)$ as desired. Similarly for the strict henselization but using Algebra, Lemma 05WS. □